

## **From Farhan Hossen:**

The **OSI Model** is a system that breaks network communication into **7 layers**, each with a specific role. It ensures that data moves smoothly from one device to another.

## **The 7 Layers of the OSI Model**

1. **Physical Layer:** Transmits raw signals (like electrical pulses or light) over cables or wireless.
2. **Data Link Layer:** Makes sure data travels without errors on a local network.
3. **Network Layer:** Decides the best path to send data between devices using IP addresses.
4. **Transport Layer:** Splits data into smaller pieces and ensures they arrive correctly.
5. **Session Layer:** Keeps track of the connection between two devices during communication.
6. **Presentation Layer:** Prepares the data for the application, handling encryption and formatting.
7. **Application Layer:** Directly interacts with users through software (like email or web browsers).

## **Example: Sending an Email (Alice to Bob)**

**Scenario:** Alice sends an email to Bob.

### **On Alice's Side (Sender):**

1. **Application Layer:** Alice writes an email in her email app. The app uses the SMTP protocol to prepare the email.
2. **Presentation Layer:** The email is converted into a format that can be sent over the network and encrypted for security.
3. **Session Layer:** A connection is created between Alice's device and her email server.
4. **Transport Layer:** The email is broken into smaller pieces (called segments). TCP ensures they are numbered correctly for reassembly.
5. **Network Layer:** Each segment is packed into a packet with Alice's and Bob's IP addresses to route it across networks.
6. **Data Link Layer:** The packet is added to a frame with hardware addresses (MAC addresses) for local delivery.

7. **Physical Layer:** The frame is converted into signals (electricity, light, or radio waves) and sent over the network.

### **On Bob's Side (Receiver):**

1. **Physical Layer:** Bob's device receives the signals and passes them to the next layer.
2. **Data Link Layer:** Frames are checked for errors and unpacked to extract packets.
3. **Network Layer:** Bob's device confirms the packet is for him (via the IP address) and forwards it.
4. **Transport Layer:** Segments are reassembled into the original email.
5. **Session Layer:** The connection between Alice's and Bob's devices remains open until the email is fully received.
6. **Presentation Layer:** The email is decrypted and formatted so it's readable.
7. **Application Layer:** Bob's email app (Gmail, Outlook) displays the email for him to read.

### **From R & D**

#### **🌐 OSI MODEL (Open Systems Interconnection Model)**

#### **1 What is OSI Model? (Definition – Exam Standard)**

**OSI Model** is a **7-layered network architecture** developed by **ISO (International Organization for Standardization)** to standardize the functions of data communication in a network.

#### **✦ বাংলায়:**

OSI Model হলো একটি **৭ স্তরবিশিষ্ট নেটওয়ার্ক মডেল**, যা **ISO** তৈরি করেছে, যাতে বিভিন্ন কম্পিউটার সিস্টেমের মধ্যে ডাটা আদান-প্রদান **সহজ, স্ট্যান্ডার্ড এবং নির্ভরযোগ্য** হয়।

#### **2 Why OSI Model is Used? (Purpose)**

OSI Model ব্যবহার করা হয় কারণ:

1. Network communication সহজভাবে বুঝতে
2. Complex network system কে layer-wise ভাগ করতে
3. Different vendor devices-এর মধ্যে compatibility নিশ্চিত করতে

4. Troubleshooting সহজ করতে
5. Network design ও development সহজ করতে

### 3 Number of Layers in OSI Model

☞ OSI Model has 7 layers

### 4 OSI Model Layers (Top to Bottom Order)

Mnemonic:

**All People Seem To Need Data Processing**

**Layer No Layer Name**

|   |              |
|---|--------------|
| 7 | Application  |
| 6 | Presentation |
| 5 | Session      |
| 4 | Transport    |
| 3 | Network      |
| 2 | Data Link    |
| 1 | Physical     |

### OSI Model Layers – A to Z (Most Important Part)

### 7 Application Layer (Layer 7)

**Definition:**

The **Application layer** provides **network services directly to the end user**.  
(Directly interacts with users through software (like email or web browsers))

**Functions:**

- User interface with network
- Email, file transfer, web access
- Provides services like HTTP, FTP

**Protocols:**

- HTTP
- FTP
- SMTP
- DNS
- Telnet

**Example:**

Opening a website using a browser

✦ **Exam line:**

Application layer is closest to the user.

## 6 Presentation Layer (Layer 6)

**Definition:**

The **Presentation layer** is responsible for **data formatting, encryption, and compression**.

(Prepares the data for the application, handling encryption and formatting.)

**Functions:**

- Data translation (format conversion)
- Encryption & decryption
- Data compression

**Example:**

- Converting data into readable format
- HTTPS encryption

✦ **Exam line:**

Presentation layer ensures data is readable and secure.

## 5 Session Layer (Layer 5)

**Definition:**

The **Session layer** establishes, manages, and terminates communication sessions.  
(Keeps track of the connection between two devices during communication.)

**Functions:**

- Session establishment
- Session maintenance
- Session termination
- Synchronization

**Example:**

- Video call session start and end

**4 Transport Layer (Layer 4) ★ ★ ★****Definition:**

The **Transport layer** ensures **end-to-end data delivery** between sender and receiver.

(Splits data into smaller pieces and ensures they arrive correctly.)

**Functions:**

- End-to-end communication
- Flow control
- Error control
- Segmentation & reassembly

**Protocols:**

- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)

**TCP vs UDP:**

| <b>TCP</b>          | <b>UDP</b>     |
|---------------------|----------------|
| Reliable            | Unreliable     |
| Connection-oriented | Connectionless |

|            |            |
|------------|------------|
| <b>TCP</b> | <b>UDP</b> |
| Slower     | Faster     |

✦ **Exam line:**

Transport layer provides reliable data transfer.

### 3 **Network Layer (Layer 3) ★ ★ ★**

**Definition:**

The **Network layer** is responsible for **routing and logical addressing**.  
(Decides the best path to send data between devices using IP addresses.)

**Functions:**

- Path determination
- Routing
- Logical addressing (IP)

**Protocols:**

- IP
- ICMP
- ARP

**Device:**

- Router

✦ **Exam line:**

Network layer decides the best path for data.

### 2 **Data Link Layer (Layer 2)**

**Definition:**

The **Data Link layer** provides **node-to-node delivery** and handles **MAC addressing**.

(Makes sure data travels without errors on a local network.)

**Functions:**

- Framing
- Error detection
- Flow control
- MAC address handling

**Device:**

- Switch

**✦ Exam line:**

Data Link layer ensures error-free transmission between two directly connected nodes.

**1 Physical Layer (Layer 1)****Definition:**

The **Physical layer** deals with **physical transmission of raw bits (0 and 1)**.  
(Transmits raw signals (like electrical pulses or light) over cables or wireless.)

**Functions:**

- Transmission of bits
- Defines cable, voltage, connectors
- Data rate control

**Devices:**

- Hub
- Cable
- Repeater

**✦ Exam line:**

Physical layer transmits raw bits over physical medium.

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## 🔗 OSI Model vs TCP/IP Model (Very Important)

| OSI Model        | TCP/IP Model       |
|------------------|--------------------|
| 7 layers         | 4 layers           |
| Conceptual model | Practical model    |
| Developed by ISO | Developed by DARPA |

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### ☐ Device & Protocol Mapping (MCQ Gold)

#### Devices:

- Hub → Physical layer
- Switch → Data Link layer
- Router → Network layer

#### Protocols:

- HTTP, FTP → Application layer
  - TCP, UDP → Transport layer
  - IP → Network layer
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### 🔗 One-Line Exam Ready Points

- Encryption → Presentation layer
- Routing → Network layer
- End-to-end delivery → Transport layer
- MAC address → Data Link layer
- Bits transmission → Physical layer

### 🔑 Final Exam Tip (Bangladesh Bank)

- ✓ Layer order
- ✓ Transport vs Network
- ✓ TCP vs UDP
- ✓ Device & protocol mapping
- ✓ OSI vs TCP/IP



□ তুমি চাইলে আমি দিতে পারি:

- ✓ Perfect 20 MCQ with answer
- ✓ 1-page OSI revision sheet
- ✓ Previous Bangladesh Bank style questions
- ✓ Memory tricks for fast recall

বললেই হবে—আমি তোমার exam partner